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Matrix solver interface mtxp in TASK

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Purpose and limitations

- **Various matrices are solved in TASK.**
 - **EQ,EQU**: Grad-Shafranov equation: R
 - **TR,TI,TX**: 1D Transport equation: R
 - **FP**: 2/3D Fokker-Planck equation: R
 - **WM,WF**: 1/2/3D fixed-frequency Maxwell's equation: C
 - **PIC**: Maxwell's equation: R
- **Various types of band matrix solver**
 - **single process** vs **MPI parallel process**
 - **iterative solver** vs **direct solver**
 - **real variables** vs **complex variables**
- **Same matrix solver interface**
- **Choice of matrix solver when compile**
- **Limitation of single choice over TASK**
 - **single or multi**,emBiterative or direct

Available matrix solver

- **single and direct**: band matrix solver
 - libmtxbnd.f90, lib/libbnd.f90
- **single and iterative**: PCGPME method
 - libmtxpcg.f90, lib/libpcgpme.f90
- **multi and direct**: MUMPS library
 - libmtxmumps.f90, DMUMPS and ZMUMPS
- **multi and iterative**: PETSC library
 - libmtxksp.f90, KSP and many others

mtxp routines

- **Routine name**
 - **mtx** for real variables, **mtxc** for complex variables
- **mtx_initialize**: Initialize
- **mtx_finalize**: Finalize
- **mtx_setup**: Define matrix size
- **mtx_set_matrix**: Set matrix element
- **mtx_set_source**: Set source vector element
- **mtx_set_vector**: Set initial solution vector for iterative scheme
- **mtx_solve**: Solve matrix equation
- **mtx_get_vector_j**: Get j -th component of solution vector
- **mtx_get_vector**: Get all components of solution vector on single CPU
- **mtx_gather_vector**: Get all components of solution vec. for multi CPU
- **mtx_cleanup**: Clear matrix data